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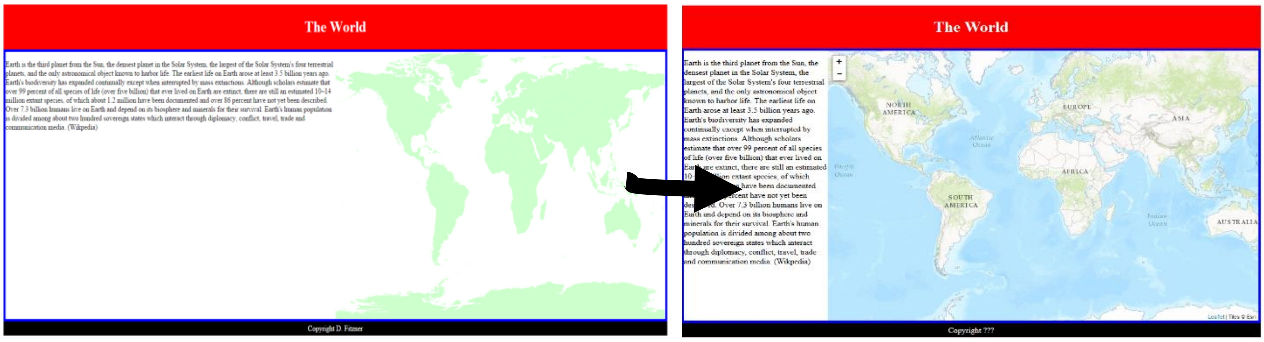
Internet-GIS WS22/23, Task 3:

Using your result from exercise 1 (HTML web page with static map image), your next task is to replace the static map image in the web page with a Leaflet map.



Task 3.1: Embed Leaflet Web Map into Website

1. Replace the static image with an area (<div>…</div>) to contain the web map. Add a new Leaflet web map to your website as shown in the lecture. Remember to import the APIs .js and .css documents. For reference, use the Leaflet API documentation (https://leafletjs.com/reference.html).
2. Select a suitable tile layer server (e.g. from https://leaflet-extras.github.io/leaflet-providers/preview/) and use it as your map background.
3. Add a scale (https://leafletjs.com/reference.html#control-scale) to your map.
4. Create a LayerGroup (https://leafletjs.com/reference.html#layergroup) to hold additional vector features to display on top of the base map (initialize the map with the background layer and the vector layer). Add a control to toggle the vector layer (https://leafletjs.com/reference.html#control-layers).
5. Add some arbitrary vector features to the vector layer, i.e. Points (https://leafletjs.com/reference.html#marker, https://leafletjs.com/reference.html#circlemarker), Lines (https://leafletjs.com/reference.html#polyline) or Polygons (https://leafletjs.com/reference.html#polygon). Use lat/lon WGS84 coordinates, obtainable using online tools, e.g. https://epsg.io/map.

Task 3.2: Map events

1. Display informations about the map on interaction using Leaflet Map Events (https://leafletjs.com/reference.html#event-objects, https://leafletjs.com/reference.html#map-event), see also https://leafletjs.com/reference.html#map-methods-for-getting-map-state:
2. Output the location of the mouse cursor on click.
3. Output the current zoom level and current visible bounds of the map after zooming in or out or moving the map.
4. Add an event listener to your vector features (events inherited from https://leafletjs.com/reference.html#interactive-layer), e.g. displaying information about the feature on click. Available layer properties can be found in the Methods-sections for each layer type.

Please upload your results to Stud.IP folder **Documents > Results of Exercise 3** until **08.11.2022**.